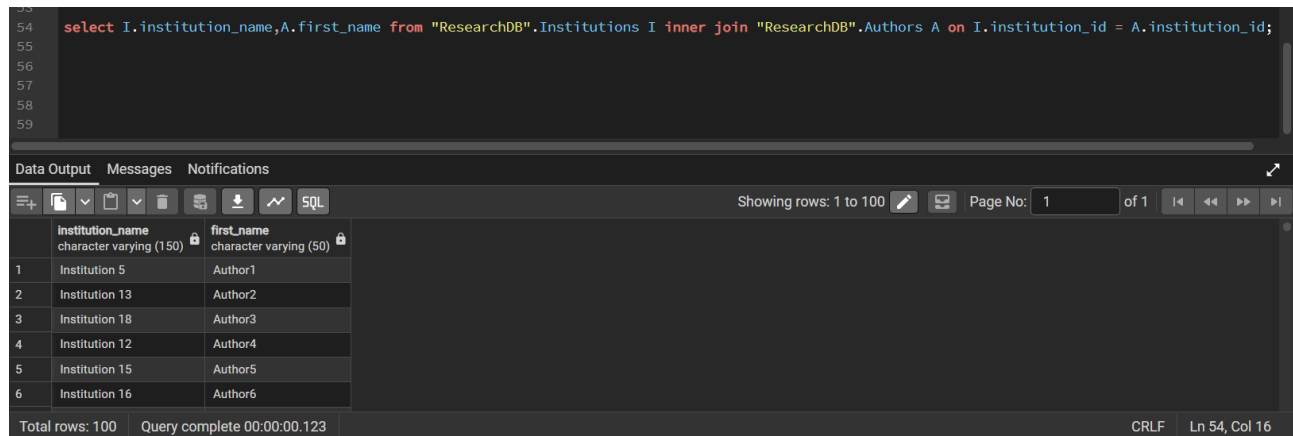


16. Display each author's institution.

```
select I.institution_name,A.first_name from "ResearchDB".Institutions I inner join  
"ResearchDB".Authors A on I.institution_id = A.institution_id;
```

Tuple count = 100



The screenshot shows a SQL query execution interface. The query is: `select I.institution_name,A.first_name from "ResearchDB".Institutions I inner join "ResearchDB".Authors A on I.institution_id = A.institution_id;`. The results are displayed in a table with two columns: `institution_name` and `first_name`. The table shows 100 rows, with the first 6 rows visible. The status bar indicates "Total rows: 100" and "Query complete 00:00:00.123".

	institution_name character varying (150)	first_name character varying (50)
1	Institution 5	Author1
2	Institution 13	Author2
3	Institution 18	Author3
4	Institution 12	Author4
5	Institution 15	Author5
6	Institution 16	Author6

17. Display paper title with conference name.

```
select p.title , c.conference_name from "ResearchDB".Papers p inner join  
"ResearchDB".Conferences c on p.conference_id = c.conference_id;
```

Tuple count = 200



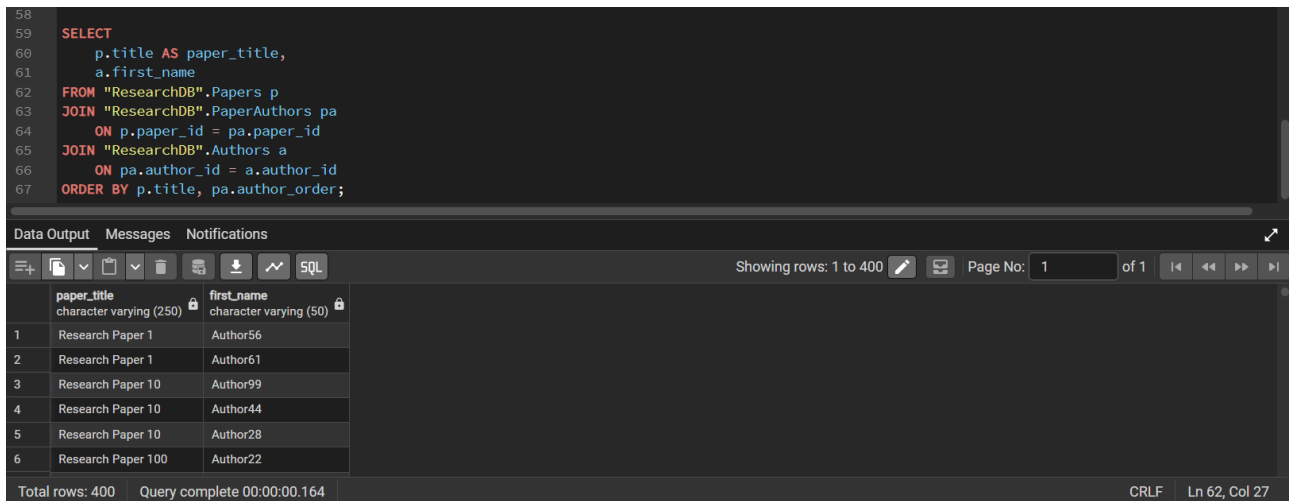
The screenshot shows a SQL query execution interface. The query is: `select p.title , c.conference_name from "ResearchDB".Papers p inner join "ResearchDB".Conferences c on p.conference_id = c.conference_id;`. The results are displayed in a table with two columns: `title` and `conference_name`. The table shows 200 rows, with the first 6 rows visible. The status bar indicates "Total rows: 200" and "Query complete 00:00:00.152".

	title character varying (250)	conference_name character varying (120)
1	Research Paper 1	Conf 2
2	Research Paper 2	Conf 1
3	Research Paper 3	Conf 16
4	Research Paper 4	Conf 5
5	Research Paper 5	Conf 18
6	Research Paper 6	Conf 5

18. Display paper title and its authors.

```
SELECT p.title AS paper_title, a.first_name FROM "ResearchDB".Papers p JOIN  
"ResearchDB".PaperAuthors pa ON p.paper_id = pa.paper_id JOIN "ResearchDB".Authors a ON  
pa.author_id = a.author_id ORDER BY p.title, pa.author_order;
```

Tuple count = 400



The screenshot shows a SQL IDE with a query editor at the top and a results pane below. The query is a JOIN of Papers, PaperAuthors, and Authors tables. The results pane shows a table with two columns: paper_title and first_name. The first six rows are visible, showing various research papers and their authors.

```
58 SELECT  
59     p.title AS paper_title,  
60     a.first_name  
61 FROM "ResearchDB".Papers p  
62 JOIN "ResearchDB".PaperAuthors pa  
63     ON p.paper_id = pa.paper_id  
64 JOIN "ResearchDB".Authors a  
65     ON pa.author_id = a.author_id  
66 ORDER BY p.title, pa.author_order;  
67
```

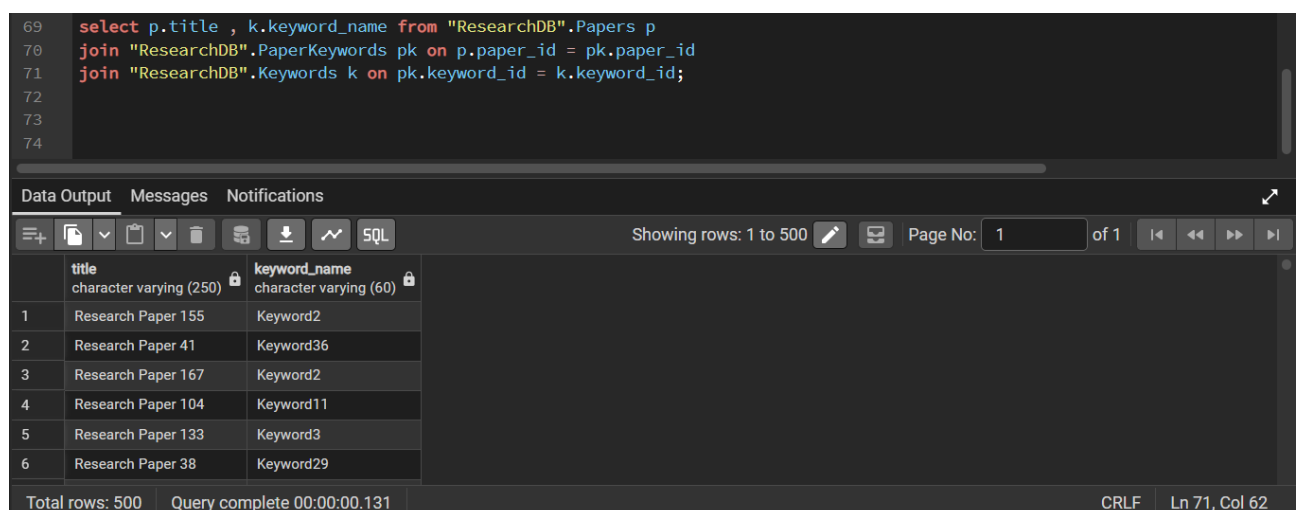
	paper_title character varying (250)	first_name character varying (50)
1	Research Paper 1	Author56
2	Research Paper 1	Author61
3	Research Paper 10	Author99
4	Research Paper 10	Author44
5	Research Paper 10	Author28
6	Research Paper 100	Author22

Total rows: 400 Query complete 00:00:00.164 CRLF Ln 62, Col 27

19 . Display papers with their keywords.

```
select p.title , k.keyword_name from "ResearchDB".Papers p join "ResearchDB".PaperKeywords  
pk on p.paper_id = pk.paper_id join "ResearchDB".Keywords k on pk.keyword_id = k.keyword_id;
```

Tuple count = 500



The screenshot shows a SQL IDE with a query editor at the top and a results pane below. The query is a JOIN of Papers, PaperKeywords, and Keywords tables. The results pane shows a table with two columns: title and keyword_name. The first six rows are visible, showing research papers and their associated keywords.

```
69 select p.title , k.keyword_name from "ResearchDB".Papers p  
70 join "ResearchDB".PaperKeywords pk on p.paper_id = pk.paper_id  
71 join "ResearchDB".Keywords k on pk.keyword_id = k.keyword_id;  
72  
73  
74
```

	title character varying (250)	keyword_name character varying (60)
1	Research Paper 155	Keyword2
2	Research Paper 41	Keyword36
3	Research Paper 167	Keyword2
4	Research Paper 104	Keyword11
5	Research Paper 133	Keyword3
6	Research Paper 38	Keyword29

Total rows: 500 Query complete 00:00:00.131 CRLF Ln 71, Col 62

20. Display author name and institution city.

```
select a.first_name , i.city from "ResearchDB".Authors a inner join "ResearchDB".Institutions i on  
a.institution_id = i.institution_id;
```

Tuple count = 100



SQL

Showing rows: 1 to 100 Page No: 1 of 1

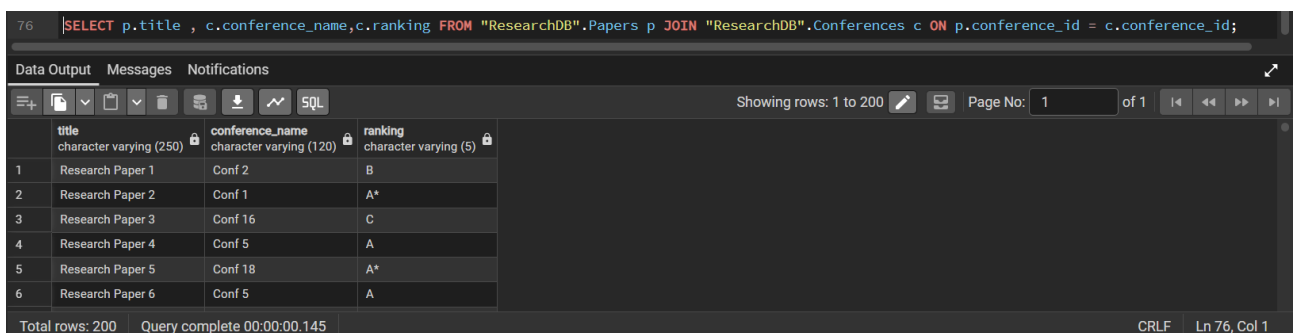
	first_name character varying (50)	city character varying (60)
1	Author1	City5
2	Author2	City13
3	Author3	City18
4	Author4	City12
5	Author5	City15
6	Author6	City16

Total rows: 100 Query complete 00:00:00.138 CRLF Ln 74, Col 136

21 . Display paper title, conference, and ranking.

```
SELECT p.title , c.conference_name,c.ranking FROM "ResearchDB".Papers p JOIN  
"ResearchDB".Conferences c ON p.conference_id = c.conference_id;
```

Tuple count = 200



SQL

Showing rows: 1 to 200 Page No: 1 of 1

	title character varying (250)	conference_name character varying (120)	ranking character varying (5)
1	Research Paper 1	Conf 2	B
2	Research Paper 2	Conf 1	A*
3	Research Paper 3	Conf 16	C
4	Research Paper 4	Conf 5	A
5	Research Paper 5	Conf 18	A*
6	Research Paper 6	Conf 5	A

Total rows: 200 Query complete 00:00:00.145 CRLF Ln 76, Col 1

22. Display all papers with citation count greater than 100 and their conference.

```
select p.title , c.conference_name , p.citation_count from "ResearchDB".Papers p inner join  
"ResearchDB".Conferences c on p.conference_id = c.conference_id where p.citation_count > 100;
```

Tuple count = 160

The screenshot shows a SQL query execution interface. The query is: `select p.title , c.conference_name , p.citation_count from "ResearchDB".Papers p inner join "ResearchDB".Conferences c on p.conference_id = c.conference_id where p.citation_count > 100;`. The results are displayed in a table with columns: title, conference_name, and citation_count. The table shows 5 rows of data, and the total row count is 160.

	title	conference_name	citation_count
71	Research Paper 86	Conf 14	143
72	Research Paper 87	Conf 13	131
73	Research Paper 88	Conf 1	450
74	Research Paper 89	Conf 6	250
75	Research Paper 90	Conf 11	427

Total rows: 160 Query complete 00:00:00.091

23.Count authors in each institution.

```
select I.institution_name , count(*) from "ResearchDB".Institutions I inner join  
"ResearchDB".Authors A on I.institution_id = A.institution_id group by I.institution_name order by  
I.institution_name;
```

Tuple count = 20

The screenshot shows a SQL query execution interface. The query is: `select I.institution_name , count(*) from "ResearchDB".Institutions I inner join "ResearchDB".Authors A on I.institution_id = A.institution_id group by I.institution_name order by I.institution_name;`. The results are displayed in a table with columns: institution_name and count. The table shows 9 rows of data, and the total row count is 20.

	institution_name	count
1	Institution 1	4
2	Institution 10	8
3	Institution 11	4
4	Institution 12	2
5	Institution 13	4
6	Institution 14	6
7	Institution 15	10
8	Institution 16	6
9	Institution 17	5

Total rows: 20 Query complete 00:00:00.105

24. Count papers in each conference.

```
select c.conference_name,count(*) from "ResearchDB".Conferences c inner join  
"ResearchDB".Papers p on p.conference_id = c.conference_id group by c.conference_id;
```

Tuple count = 20

```
63 select c.conference_name,count(*) from "ResearchDB".Conferences c inner join "ResearchDB".Papers p
64 on p.conference_id = c.conference_id group by c.conference_id;
```

	conference_name character varying (120)	count bigint
1	Conf 11	8
2	Conf 9	9
3	Conf 15	13
4	Conf 19	10
5	Conf 3	9
6	Conf 17	10
7	Conf 5	9
8	Conf 4	8
9	Conf 10	10

Total rows: 20 Query complete 00:00:00.140 CRLF Ln 63, Col 50

25. Display conferences having more than 10 papers.

```
select c.conference_name,count(p.paper_id) from "ResearchDB".Conferences c inner join  
"ResearchDB".Papers p on c.conference_id = p.conference_id group by c.conference_name having  
count(p.paper_id) > 10;
```

Tuple count = 6

```
66 select c.conference_name,count(p.paper_id) from "ResearchDB".Conferences c inner join "ResearchDB".Papers p
67 on c.conference_id = p.conference_id group by c.conference_name having count(p.paper_id) > 10;
```

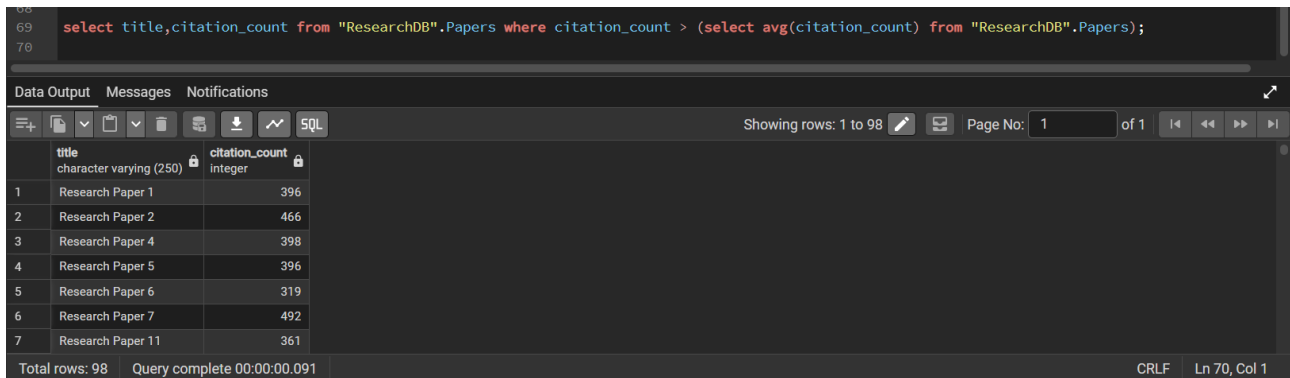
	conference_name character varying (120)	count bigint
1	Conf 12	15
2	Conf 1	13
3	Conf 8	13
4	Conf 15	13
5	Conf 14	11
6	Conf 13	13

Total rows: 6 Query complete 00:00:00.091 CRLF Ln 67, Col 1

26. Find papers having citation count greater than the average.

```
select title,citation_count from "ResearchDB".Papers where citation_count > (select avg(citation_count) from "ResearchDB".Papers);
```

Tuple count = 98



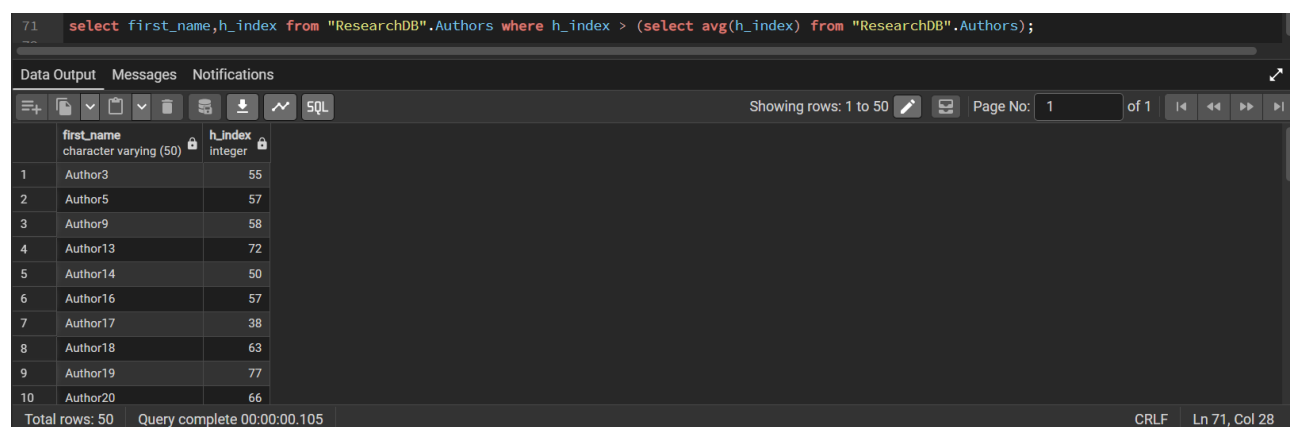
The screenshot shows a SQL query execution interface. The query is: `select title,citation_count from "ResearchDB".Papers where citation_count > (select avg(citation_count) from "ResearchDB".Papers);`. The results are displayed in a table with two columns: `title` (character varying (250)) and `citation_count` (integer). The table shows 7 rows of data, representing a subset of the 98 total rows. The status bar indicates "Total rows: 98" and "Query complete 00:00:00.091".

	title	citation_count
1	Research Paper 1	396
2	Research Paper 2	466
3	Research Paper 4	398
4	Research Paper 5	396
5	Research Paper 6	319
6	Research Paper 7	492
7	Research Paper 11	361

27. Find authors whose h-index is above the average.

```
select first_name,h_index from "ResearchDB".Authors where h_index > (select avg(h_index) from "ResearchDB".Authors);
```

Tuple count = 50



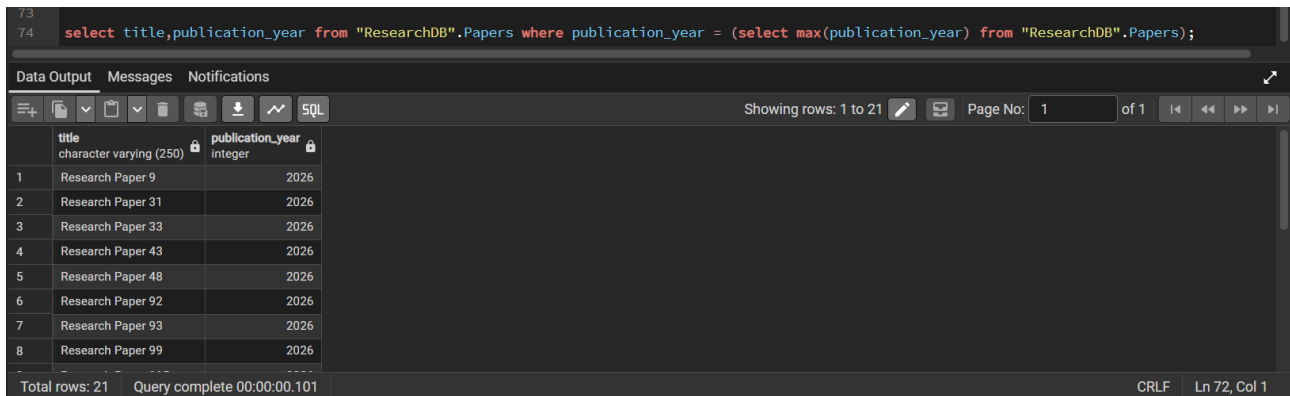
The screenshot shows a SQL query execution interface. The query is: `select first_name,h_index from "ResearchDB".Authors where h_index > (select avg(h_index) from "ResearchDB".Authors);`. The results are displayed in a table with two columns: `first_name` (character varying (50)) and `h_index` (integer). The table shows 10 rows of data, representing a subset of the 50 total rows. The status bar indicates "Total rows: 50" and "Query complete 00:00:00.105".

	first_name	h_index
1	Author3	55
2	Author5	57
3	Author9	58
4	Author13	72
5	Author14	50
6	Author16	57
7	Author17	38
8	Author18	63
9	Author19	77
10	Author20	66

28. Display papers published in the latest publication year.

```
select title,publication_year from "ResearchDB".Papers where publication_year = (select max(publication_year) from "ResearchDB".Papers);
```

Tuple count = 21



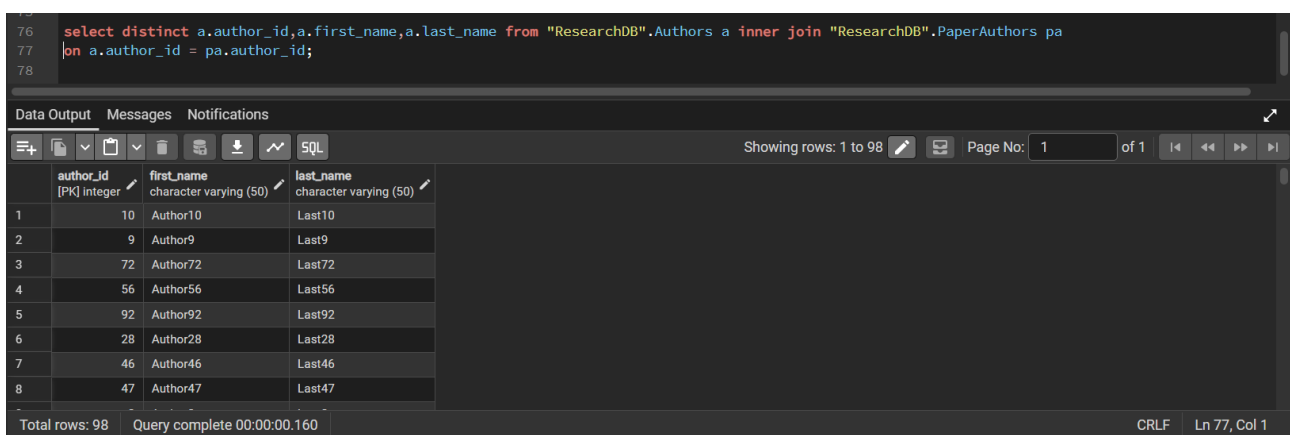
	title character varying (250)	publication_year integer
1	Research Paper 9	2026
2	Research Paper 31	2026
3	Research Paper 33	2026
4	Research Paper 43	2026
5	Research Paper 48	2026
6	Research Paper 92	2026
7	Research Paper 93	2026
8	Research Paper 99	2026
...	...	...

Total rows: 21 Query complete 00:00:00.101 CRLF Ln 72, Col 1

29. Display authors who have written at least one paper.

```
select distinct a.author_id,a.first_name,a.last_name from "ResearchDB".Authors a inner join "ResearchDB".PaperAuthors pa on a.author_id = pa.author_id;
```

Tuple count = 98



	author_id [PK] integer	first_name character varying (50)	last_name character varying (50)
1	10	Author10	Last10
2	9	Author9	Last9
3	72	Author72	Last72
4	56	Author56	Last56
5	92	Author92	Last92
6	28	Author28	Last28
7	46	Author46	Last46
8	47	Author47	Last47
...	...	...	...

Total rows: 98 Query complete 00:00:00.160 CRLF Ln 77, Col 1

30. Create a view to display paper and conference details.

create view paper_conference as select p.paper_id , p.title , p.publication_year ,
p.citation_count,c.conference_name, c.conference_type,c.publisher,c.ranking from
"ResearchDB".Papers p join "ResearchDB".Conferences c on p.conference_id = c.conference_id;

```
79 create view paper_conference as
80 select p.paper_id , p.title , p.publication_year , p.citation_count,c.conference_name, c.conference_type,c.publisher,c.ranking
81 from "ResearchDB".Papers p join "ResearchDB".Conferences c on p.conference_id = c.conference_id;
```

Data Output Messages Notifications

CREATE VIEW

Query returned successfully in 160 msec.

Total rows: Query complete 00:00:00.160 CRLF Ln 81, Col 98

```
82
83 select * from paper_conference;
```

Data Output Messages Notifications

Showing rows: 1 to 200 Page No: 1 of 1

	paper_id integer	title character varying (250)	publication_year integer	citation_count integer	conference_name character varying (120)	conference_type character varying (20)	publisher character varying (80)	ranking character varying (5)
1	1	Research Paper 1	2022	396	Conf 2	Journal	IEEE	B
2	2	Research Paper 2	2021	466	Conf 1	Conference	IEEE	A*
3	3	Research Paper 3	2023	188	Conf 16	Journal	IEEE	C
4	4	Research Paper 4	2018	398	Conf 5	Journal	IEEE	A
5	5	Research Paper 5	2019	396	Conf 18	Journal	Springer	A*
6	6	Research Paper 6	2023	319	Conf 5	Journal	IEEE	A
7	7	Research Paper 7	2020	492	Conf 13	Journal	Springer	A

Total rows: 200 Query complete 00:00:00.156 CRLF Ln 83, Col 32